Lecture 3

Tuesday, 29 August 2023 10:33 AM

Infra-red spectroscopy

+ Identity of adsorbed species
+ Surface structures and phases
+ In-situ & operando techniques

Usually too stable to be an intermediate species

CO → 2143 cm^{-1} (free)

Frequency shifts & BE:

Stronger bond → Less energy to vibrate that bond → shift

↳ Antibonding

Backbond → need e^-

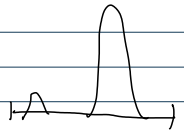
↑ wavenu. → stronger ↑ ???

$E = \frac{hc}{\lambda}$ $\frac{1}{\lambda}$ = wavenu.

$E = hc \cdot$ wave. no.

New MO formed when hy → M d orbitals interacting w/ π orbitals in CO → It can happen that σ bond could be de-stabilized more than the π bonding

CO on Pt(111)



+ Area increase (due to coverage)
+ Shift due to interactions

Operando transmission IR:

peak area correlates w/ time ; path length of gas should be minimized

Under reaction conditions

→ Add the tracer to know hydrodynamic vs. reaction kinetics

IR feature corresponding w/ something intermediate
↳ that is the intermediate

Intrinsic TOF → normalized TOF to something

to something

Steady state Isotopic Transient Kinetic Analysis (SSITKA)

The time constant for hydrodynamic (transport) constrictions is given by the carrier MS signal